

IBM Bob Pull Request Review Report

Project: PR Guardian AI

Hackathon: IBM TechXchange 2026 Pre-Conference Dev Day

Repository: megandas/pr-guardian-ai-ibmbob

Generated By: IBM Bob AI Review Engine (IBM Granite)

Metric	Result
PR Health Score	92 / 100
Merge Status	READY TO MERGE
Security Risk	Low (after recommendations)
Architecture Coverage	Frontend + Backend + RAG + IBM Granite

Feature Summary

Replay AI Review, Interactive Code Impact Map, Repository Intelligence Dashboard, IBM Bob Chat, PR Document Understanding.

Developer Intent

Improve pull request reviews by combining IBM Granite reasoning, security analysis, testing recommendations, and repository intelligence inside IBM Bob IDE.

Security Review Findings

Avoid hardcoded credentials. Use environment variables. Validate repository inputs. Keep API keys out of GitHub.

Testing Suggestions

Authentication tests, security validation, Replay AI rendering, dashboard UI tests, FastAPI endpoint integration.

Repository Intelligence

Frontend extension communicates with FastAPI backend, which invokes IBM Granite and RAG services for repository-aware analysis.

Replay AI Workflow

Repository parsing → Security Agent → Repository Intelligence → Testing Agent → Merge Recommendation.

Merge Readiness

Recommendation: READY TO MERGE after regression testing. Repository architecture is modular and follows IBM hackathon best practices.

IBM Technologies Used

IBM watsonx.ai, IBM Granite 4 Small, IBM Bob IDE, FastAPI, LangChain, ChromaDB, TypeScript, Python.

Submission Notes

This PDF is included in the GitHub repository as the exported IBM Bob report required for the IBM TechXchange Hackathon submission. Screenshots of the Repository Dashboard, Replay AI Review, Code Impact Map, and PR Document Understanding should be placed in the docs/ folder alongside this report.